

Generation Interconnection System Impact Study Report

For Queue Project Q17-0815

Devilliers 138 kV
February 15, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0815. The Interconnection Customer proposes a 117.2 MW battery energy storage facility interconnecting at the Devilliers 138 kV substation in the West Virginia area. The request was received on June 26, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 289 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 37 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$89,306,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Rio Grande - Mogadore 138 KV
Contingency: Scarbro - Wastewater 138 KV line
Loading: 143.4% (Rating: 246 MVA)
DFAX: 10.4%
Pre-project loading: 138.4%
2. Facility: Beury Mountain - Rio Grande 138 KV
Contingency: Scarbro - Wastewater 138 KV
Loading: 142.3% (Rating: 251 MVA)
DFAX: 10.4%
Pre-project loading: 137.4%
3. Facility: Productos Electricos 2 - Beury Mtn 138 KV L
Contingency: Scarbro - Wastewater 138 KV
Loading: 139.2% (Rating: 265 MVA)
DFAX: 10.4%
Pre-project loading: 134.6%
4. Facility: Oyster Bay - Osceola Solar Facility 138 KV
Contingency: Wastewater - Scarbro 138 KV Line
Loading: 137.9% (Rating: 272 MVA)
DFAX: 10.4%
Pre-project loading: 133.4%
5. Facility: Productos Electricos 2 - Oyster Bay 138 KV Line
Contingency: Wastewater - Scarbro 138 KV
Loading: 137.6% (Rating: 270 MVA)
DFAX: 10.4%
Pre-project loading: 133.1%
6. Facility: Vergennes 9 Substation - E Holbrook 138 KV Line
Contingency: Wastewaer - Scarbro 138 KV L
Loading: 137.0% (Rating: 277 MVA)

DFAX: 10.4%

Pre-project loading: 132.6%

7. Facility: Gears - Chief Joseph 138 KV

Contingency: Beech Ridge Energy Llc - Big Basin 138 KV Line

Loading: 135.6% (Rating: 207 MVA)

DFAX: 9.2%

Pre-project loading: 130.4%

8. Facility: Beaver County - Frio Town 138 KV

Contingency: Batesville - Devilliers Substation 138 KV L

Loading: 131.7% (Rating: 176 MVA)

DFAX: 50.3%

Pre-project loading: 98.3%

9. Facility: East Holbrook Substation - Osceola Solar Facility 138 KV

Contingency: Scarbro - Wastewater 138 KV

Loading: 131.0% (Rating: 288 MVA)

DFAX: 10.4%

Pre-project loading: 126.8%

10. Facility: Donnells-Curtis - Chief Joseph 138 KV

Contingency: Big Basin - Beech Ridge Energy Llc 138 KV

Loading: 128.5% (Rating: 234 MVA)

DFAX: 7.6%

Pre-project loading: 124.6%

11. Facility: Donnells-Curtis - Beech Ridge Energy Llc 138 KV L

Contingency: Beech Ridge Energy Llc - Big Basin 138 KV

Loading: 128.1% (Rating: 236 MVA)

DFAX: 7.6%

Pre-project loading: 124.3%

12. Facility: American Falls - Falconer 138 KV

Contingency: Scarbro - N Sweetwater 138 KV

Loading: 124.9% (Rating: 223 MVA)

DFAX: 7.5%

Pre-project loading: 121.0%

13. Facility: Frio Town - Heyburn Substation 138 KV

Contingency: Batesville - Devilliers 138 KV

Loading: 122.2% (Rating: 195 MVA)

DFAX: 50.3%

Pre-project loading: 92.0%

14. Facility: Tilton - Gears 138 KV

Contingency: Brockway - Gears 138 KV line

Loading: 119.8% (Rating: 202 MVA)

DFAX: 6.1%

Pre-project loading: 116.3%

15. Facility: Pax Branch - Falconer 138 KV

Contingency: Scarbro - N Sweetwater 138 KV

Loading: 119.3% (Rating: 212 MVA)

DFAX: 7.5%

Pre-project loading: 115.2%

16. Facility: Meadow Bridge - Beaver Conty 138 KV

Contingency: Batesville - Devilliers 138 KV Line

Loading: 115.9% (Rating: 196 MVA)

DFAX: 50.3%

Pre-project loading: 85.8%

17. Facility: Brockway - Gears 138 KV

Contingency: Gears - Tilton 138 KV

Loading: 115.0% (Rating: 202 MVA)

DFAX: 5.9%

Pre-project loading: 111.5%

18. Facility: Falls City - Berlin Ctr 138 KV

Contingency: Hinton Ap - Antelope Vly Sta 138 KV

Loading: 114.2% (Rating: 187 MVA)

DFAX: 7.3%

Pre-project loading: 109.7%

19. Facility: Lysander - Heyburn 138 KV

Contingency: Scarbro - Wastewater 138 KV Line

Loading: 112.3% (Rating: 324 MVA)

DFAX: 31.1%

Pre-project loading: 101.0%

20. Facility: Lysander - Raver 138 KV

Contingency: Wastewater - Scarbro 138 KV

Loading: 111.8% (Rating: 335 MVA)

DFAX: 31.1%

Pre-project loading: 100.9%

21. Facility: Sherman Road - Raver 138 KV

Contingency: Scarbro - Wastewater 138 KV line

Loading: 111.6% (Rating: 338 MVA)

DFAX: 31.1%

Pre-project loading: 100.8%

22. Facility: Pax Branch - Berlin 138 KV Line

Contingency: North Sweetwater - Scarbro 138 KV

Loading: 110.6% (Rating: 223 MVA)

DFAX: 7.5%

Pre-project loading: 106.7%

23. Facility: Oso - Brockway 138 KV Line

Contingency: Tilton Substation - Gears 138 KV

Loading: 109.8% (Rating: 208 MVA)

DFAX: 5.9%

Pre-project loading: 106.5%

24. Facility: Berlin - Scarbro 138 KV

Contingency: Scarbro - N Sweetwater 138 KV

Loading: 108.1% (Rating: 226 MVA)

DFAX: 7.5%

Pre-project loading: 104.2%

25. Facility: Antelope Valley Sta - Meadow Bridge 138 KV

Contingency: Devilliers - Batesville 138 KV

Loading: 108.0% (Rating: 333 MVA)

DFAX: 33.5%

Pre-project loading: 96.2%

26. Facility: Hawks Nest Hydro - Shingle Springs 138 KV

Contingency: Saltery Bay - Gauley Mtn 138 KV

Loading: 107.2% (Rating: 517 MVA)

DFAX: 9.5%

Pre-project loading: 105.0%

27. Facility: Alpha Rd - Chief Joseph 138 KV
Contingency: Vergennes 9 - Beech Ridge Energy Llc 345 KV Line
Loading: 106.8% (Rating: 202 MVA)
DFAX: 10.2%
Pre-project loading: 100.8%
28. Facility: Falls City - Harriman 138 KV
Contingency: Hinton Ap - Antelope Valley Station 138 KV
Loading: 106.6% (Rating: 194 MVA)
DFAX: 7.3%
Pre-project loading: 102.2%
29. Facility: Neosho - Beardsley St 138 KV
Contingency: Lowmoor - Hinton Ap 345 KV
Loading: 106.1% (Rating: 204 MVA)
DFAX: 6.0%
Pre-project loading: 102.7%
30. Facility: Alpha Rd - Pawnee 138 KV
Contingency: Beech Ridge Energy Llc 138/1 kV TX
Loading: 105.7% (Rating: 332 MVA)
DFAX: 14.0%
Pre-project loading: 100.8%
31. Facility: Sherbino I - Harriman 138 KV Line
Contingency: Hinton Ap - Antelope Valley Station 138 KV
Loading: 105.1% (Rating: 195 MVA)
DFAX: 7.3%
Pre-project loading: 100.7%
32. Facility: Vaseaux Lake - Red Hills 138 KV
Contingency: Lowmoor - Hinton Ap Substation 345 KV
Loading: 104.4% (Rating: 415 MVA)
DFAX: 10.4%
Pre-project loading: 101.5%
33. Facility: McClain Energy Facility Substation - Vaseaux Lake 138 KV
Contingency: Hinton Ap - Lowmoor 345 KV
Loading: 104.4% (Rating: 418 MVA)
DFAX: 10.4%
Pre-project loading: 101.5%
34. Facility: Devilliers - Batesville 138 KV
Contingency: Heyburn - Frio Town 138 KV
Loading: 104.1% (Rating: 247 MVA)
DFAX: 63.6%
Pre-project loading: 73.9%
35. Facility: Sherbinol - Coal 138 KV
Contingency: Antelope Valley Sta - Hnton Ap 138 KV
Loading: 101.9% (Rating: 199 MVA)
DFAX: 7.3%
Pre-project loading: 97.6%
36. Facility: Beech Ridge Energy Llc 345/1kV TX
Contingency: Vergennes 9 345/138kV Auto
Loading: 100.0% (Rating: 565 MVA)
DFAX: 16.1%
Pre-project loading: 96.7%
37. Facility: Beech Ridge Energy Llc 138 kV/1 kV Transformer Bank
Contingency: Vergennes 9 345/138 kV TX

Loading: 100.0% (Rating: 565 MVA)
 DFAX: 16.1%
 Pre-project loading: 96.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rio Grande - Mogadore 138 KV	Rebuild line with higher-capacity conductor	\$2,702,000	16% (\$427,000)
Beury Mountain - Rio Grande 138 KV	Rebuild line with higher-capacity conductor	\$17,027,000	16% (\$2,690,000)
Productos Electricos 2 - Beury Mtn 138 KV L	Rebuild line with higher-capacity conductor	\$9,965,000	16% (\$1,574,000)
Osceola Solar Facility - Oyster Bay 138 KV line	Rebuild line with higher-capacity conductor	\$8,548,000	16% (\$1,351,000)
Productos Electricos 2 - Oyster Bay 138 KV Line	Rebuild line with higher-capacity conductor	\$9,965,000	16% (\$1,574,000)
Vergennes 9 Substation - E Holbrook 138 KV Line	Rebuild line with higher-capacity conductor	\$6,693,000	16% (\$1,058,000)
Gears - Chief Joseph 138 KV	Rebuild line with higher-capacity conductor	\$17,082,000	11% (\$1,865,000)
Beaver County - Frio Town 138 KV	Rebuild line with higher-capacity conductor	\$16,113,000	100% (\$16,113,000)
East Holbrook Substation - Osceola Solar Facility 138 KV	Rebuild line with higher-capacity conductor	\$7,372,000	16% (\$1,165,000)
Donnells-Curtis - Chief Joseph 138 KV	Rebuild line with higher-capacity conductor	\$3,854,000	7% (\$275,000)
Donnells-Curtis - Beech Ridge Energy Llc 138 KV	Rebuild line with higher-capacity conductor	\$8,367,000	7% (\$597,000)
American Falls - Falconer 138 KV	Rebuild line with higher-capacity conductor	\$3,809,000	100% (\$3,809,000)
Frio Town - Heyburn Substation 138 KV	Rebuild line with higher-capacity conductor	\$8,535,000	100% (\$8,535,000)
Tilton - Gears 138 KV	Rebuild line with higher-capacity conductor	\$29,044,000	5% (\$1,510,000)
Pax Branch - Falconer 138 KV	Rebuild line with higher-capacity conductor	\$3,213,000	100% (\$3,213,000)
Meadow Bridge - Beaver Conty 138 KV	Rebuild line with higher-capacity conductor	\$22,359,000	100% (\$22,359,000)
Brockway - Gears 138 KV	Rebuild line with higher-capacity conductor	\$33,284,000	5% (\$1,748,000)
Berlin Ctr - Falls City 138 KV L	Rebuild line with higher-capacity conductor	\$6,051,000	9% (\$546,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lysander - Heyburn 138 KV	Rebuild line with higher-capacity conductor	\$21,983,000	27% (\$5,968,000)
Lysander - Raver 138 KV	Rebuild line with higher-capacity conductor	\$5,748,000	27% (\$1,560,000)
Sherman Road - Raver 138 KV	Rebuild line with higher-capacity conductor	\$4,612,000	27% (\$1,252,000)
Pax Branch - Berlin 138 KV Line	Rebuild line with higher-capacity conductor	\$3,355,000	100% (\$3,355,000)
Oso - Brockway 138 KV Line	Reconductor line	\$1,335,000	5% (\$70,000)
Scarbro - Berlin 138 KV	Reconductor line	\$1,117,000	100% (\$1,117,000)
Meadow Bridge - Antelope Vly Sta Substation 138 KV	Reconductor line	\$3,678,000	9% (\$331,000)
Hawks Nest Hydro - Shingle Springs 138 KV	Reconductor line	\$1,066,000	6% (\$61,000)
Alpha Rd - Chief Joseph 138 KV	Reconductor line	\$1,561,000	100% (\$1,561,000)
Falls City - Harriman 138 KV	Reconductor line	\$1,468,000	9% (\$132,000)
Neosho - Beardsley St 138 KV L	Reconductor line	\$1,725,000	15% (\$257,000)
Alpha Road - Pawnee 138 KV	Reconductor line	\$1,917,000	11% (\$204,000)
Harriman - Sherbino I 138 KV Line	Reconductor line	\$1,738,000	9% (\$157,000)
Red Hills - Vaseaux Lake 138 KV	Reconductor line	\$5,251,000	3% (\$164,000)
Vaseaux Lake - McClain Energy Facility 138 KV	Reconductor line	\$5,905,000	3% (\$184,000)
Devilliers - Batesville 138 KV	Reconductor line	\$2,128,000	53% (\$1,136,000)
Coal - Sherbino I 138 KV	Reconductor line	\$774,000	9% (\$70,000)
Beech Ridge Energy Llc 345/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$16,730,000	6% (\$1,030,000)
Beech Ridge Energy Llc 138 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$4,670,000	6% (\$288,000)
	Total allocated to project		\$89,306,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.